# Bayesian Classification

# Performance Analysis

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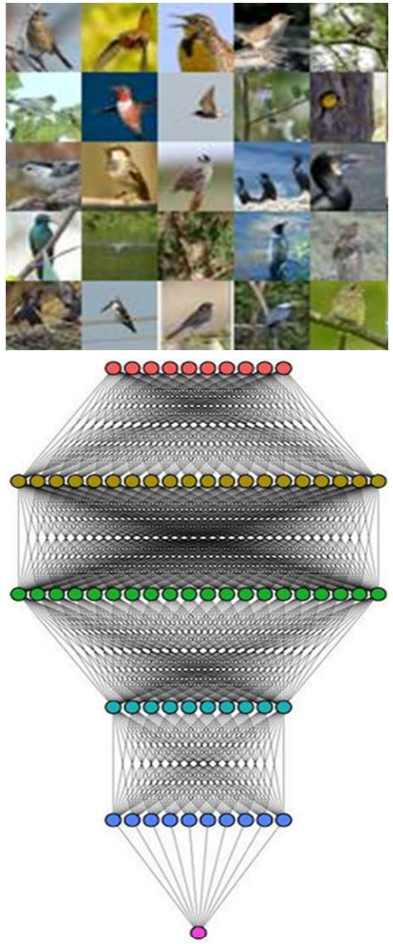
# Supervised Learning



- Bayesian Classification, MAP, Chebyshev Inequality
- Performance Measures, Confusion Matrix, ROC Curves
- Logistic Regression
- Perceptron
- Multi-Layer Perceptron (MLP), ELM
- MLP Architectures, Learning, Interpretations
- Non-parametric Methods and K-NN
- Radial Basis Function Neural Networks
- Data Balancing; SMOTE & Weighted Loss Functions
- Classification & Regression Trees
- Support Vector Machines & Multiple Kernel Learning
- Ensemble Methods, Bagging and Boosting

# Classification using Chebyshev Inequality

$$S = \{(\boldsymbol{x}_i, y_i); i = 1, \dots n\}$$

$\boldsymbol{x} \in \mathbb{R}^d$

$y \in \{0,1\}$

# Mathematical Expectation: Recapitulation

Consider Random Variable $x$ with Density/Mass Function $p(x)$

$$E[g(X)] = \int_{-\infty}^{\infty} g(X)p(X)dX \qquad E[h(X)] = \sum_i h(X_i)p(X_i)$$

Linear Operator: $E[a_1 g(X) + a_2 h(X)] = a_1 E[g(X)] + a_2 E[h(X)]$

Operating on a Constant K: $E[K] = K$

$E[X]: Mean$

$E[(X-\mu)^2]: Variance$

## Example: Random Variable $X$ with Poisson P.M.F.

$$P(x;\lambda) = \frac{e^{-\lambda}\lambda^{x}}{x!}$$

$$\mu_X = E(X) = ?$$

$$Var(X) = E[(X - \mu_X)^2] = ?$$

# Example: Random Variable $X$ with Poisson P.M.F.

$$P(x;\lambda) = \frac{e^{-\lambda}\lambda^x}{x!} \quad \Rightarrow \quad E(X) = \sum_{x=0}^{\infty} x\, P(x;\lambda) = \sum_{x=0}^{\infty} x \frac{e^{-\lambda}\lambda^x}{x!}$$

$$\sum_{x=0}^{\infty} x \frac{e^{-\lambda}\lambda^x}{x!} = e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda \cdot \lambda^{(x-1)}}{(x-1)!} = \lambda e^{-\lambda} \sum_{r=0}^{\infty} \frac{\lambda^r}{r!} = \lambda$$

$$\boxed{E(X) = \mu_X = \lambda}$$

# Example: Random Variable $X$ with Poisson P.M.F.

$$Var(X) = E[(X - \mu_X)^2] = E[X^2 - 2X\mu_X + \mu_X^2]$$

$$Var(X) = E(X^2) - 2\mu_X E(X) + E(\mu_X^2) = E(X^2) - \mu_X^2$$

$$E(X^2) = \sum_{x=0}^{\infty} x^2\, p(x) = \sum_{x=0}^{\infty} x(x-1)p(x) + \sum_{x=0}^{\infty} xp(x)$$

# Example: Random Variable $X$ with Poisson P.M.F.

$$\sum_{x=0}^{\infty} x(x-1)p(x) = \sum_{x=0}^{\infty} x(x-1)\frac{e^{-\lambda}\lambda^{x}}{x!} = e^{-\lambda}\sum_{x=2}^{\infty}\frac{\lambda^{2}\cdot\lambda^{(x-2)}}{(x-2)!} = \lambda^{2}$$

$$\sum_{x=0}^{\infty} xp(x) = \lambda \qquad Var(X) = E(X^{2}) - \mu_{X}^{2} = \lambda^{2} + \lambda - \lambda^{2}$$

$$\boxed{Var(X) = \lambda}$$

# Markov Inequality

Consider the Random Variable $Y \geq 0$ and a Value $a > 0$

$$P\{Y \geq \mathrm{a}\} \leq \frac{E[Y]}{a}$$

# Chebyshev Inequality

Substitutions $\longrightarrow$ $a = \lambda^2; \quad \lambda > 0$

$$Y = \left(\frac{X - \mu_X}{\sigma_X}\right)^2 \longrightarrow \sigma_X^2 = E[(X - \mu_X)^2]$$

Random Variable $X \in \mathbb{R}^1$

$$\mu_X = E[X]$$

$$E[Y] = ?$$

# Chebyshev Inequality

Substitutions $\longrightarrow$ $a = \lambda^2; \quad \lambda > 0$

$$Y = \left(\frac{X - \mu_X}{\sigma_X}\right)^2$$

$$\mu_X = E[X]$$

$$\sigma_X^2 = E[(X - \mu_X)^2]$$

Random Variable $X \in \mathbb{R}^1$

$$E[Y] = \frac{E[(X - \mu_X)^2]}{\sigma_X^2} = 1$$

# Chebyshev Inequality

## Substituting in Markov Inequality

$$P\left\{\left(\frac{X-\mu_X}{\sigma_X}\right)^2 \geq \lambda^2\right\} \leq \frac{E[Y]}{\lambda^2}$$

$$\Downarrow$$

$$P\{|X-\mu_X| \geq \lambda\sigma_X\} \leq \frac{1}{\lambda^2}$$

# Chebyshev Inequality

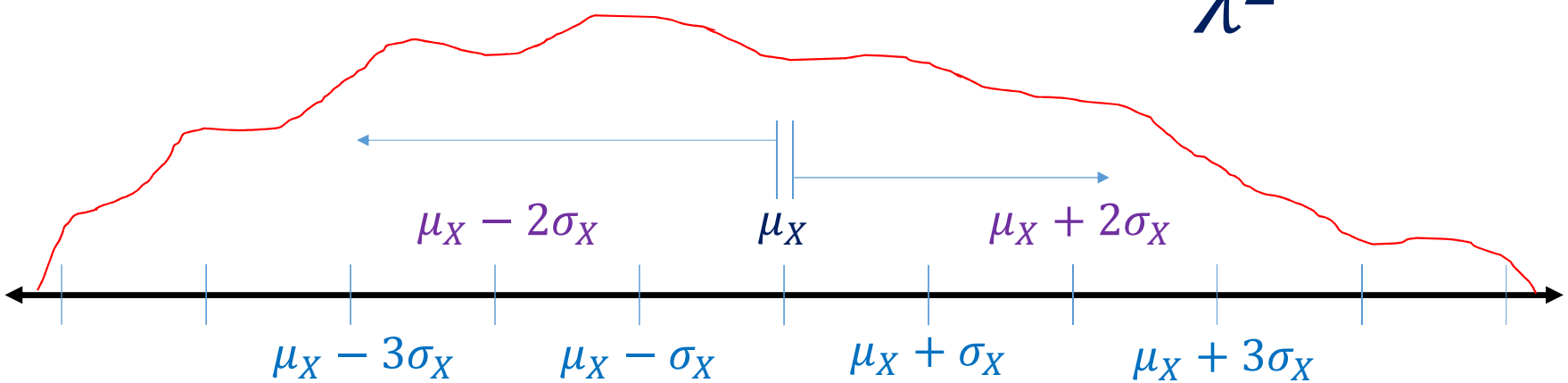
Provides an Upper Bound on Membership for an Instance based on its Distance from Mean Normalized by Standard Deviation. No Properties of the Underlying Distribution are Assumed.

$$P\{|X - \mu_X| \geq \lambda\sigma_X\} \leq \frac{1}{\lambda^2}$$

# Chebyshev Inequality

Provides an Upper Bound on Membership for an Instance based on its Distance from Mean Normalized by Standard Deviation. No Properties of the Underlying Distribution are Assumed.

$$P\{|X - \mu_X| \geq \lambda\sigma_X\} \leq \frac{1}{\lambda^2}$$



# Chebyshev Inequality as Classifier

Training Data of Class $\omega$: $S = \{x_i; i = 1, \dots n; x \in \mathbb{R}^1\}$

$$\mu = \sum_{i=1}^{n} x_i$$

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2$$

Classify Unseen Data: $u$

Classification Rule: $u \in \omega \Rightarrow |u - \mu| \leq \lambda\sigma$

---

Training Data of Class $\omega$: $S = \{2\,, 3\,, 5\,, -1\,, 10\,, -3\,, 6\,, 11\}$

$$\mu = 4.125$$

$$\sigma = 4.5945$$

Classify Unseen Data: $u = 16$ using $\lambda = 3$

Range for Class $\omega$: $[\mu - \lambda\sigma\,,\ \mu + \lambda\sigma\,] = [-9.6585\,, 17.9085]$

$u = 16$ Lies in the Interval $[-9.6585\,, 17.9085]$ and Hence belongs to Class $\omega$

# Classification for Multivariate Data

Training Data of Class $\boldsymbol{\omega}$: $\boldsymbol{S} = \{\boldsymbol{x}_i; i = 1, \dots n; \boldsymbol{x} \in \mathbb{R}^{\boldsymbol{d}}\}$

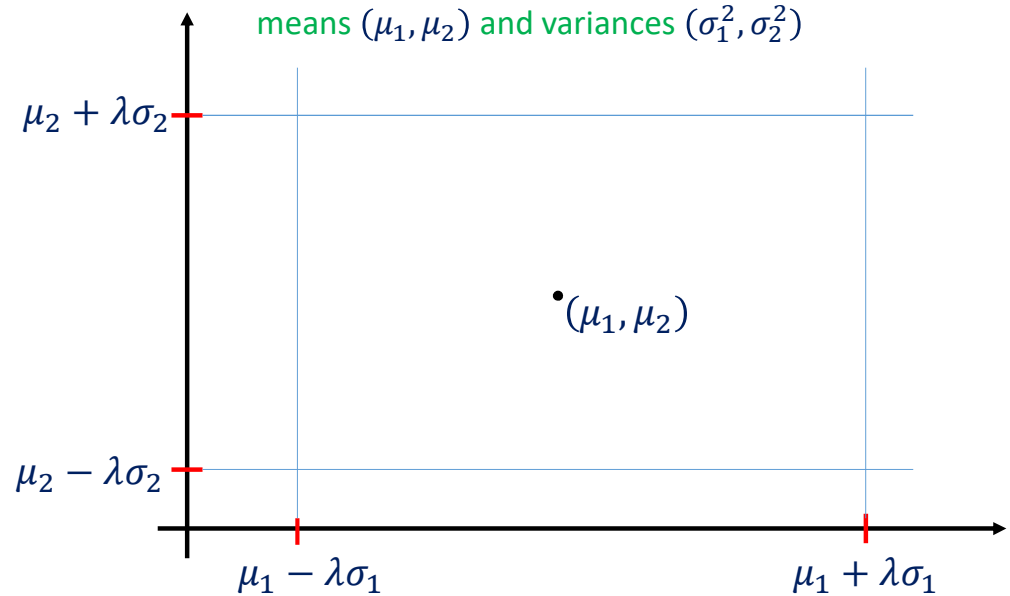
$$\mu_j = \sum_{i=1}^{n} \boldsymbol{x}_i[j] \qquad j = 1, \dots d \qquad v_j = \frac{1}{n}\sum_{i=1}^{n}\left(\boldsymbol{x}_i[j] - \mu_j\right)^2$$

Classification Rule $\boldsymbol{R}_k$ along Dimension $k$ have Boolean Values

$$\boldsymbol{R}_k = \begin{cases} TRUE, (\boldsymbol{x}[k] - \mu_k)^2 \leq \lambda^2 v_k \\ FALSE, (\boldsymbol{x}[k] - \mu_k)^2 > \lambda^2 v_k \end{cases}$$

$$\boldsymbol{x} \in \boldsymbol{\omega} \Rightarrow \left[\bigwedge_{k=1}^{d} \boldsymbol{R}_k = TRUE\right]$$

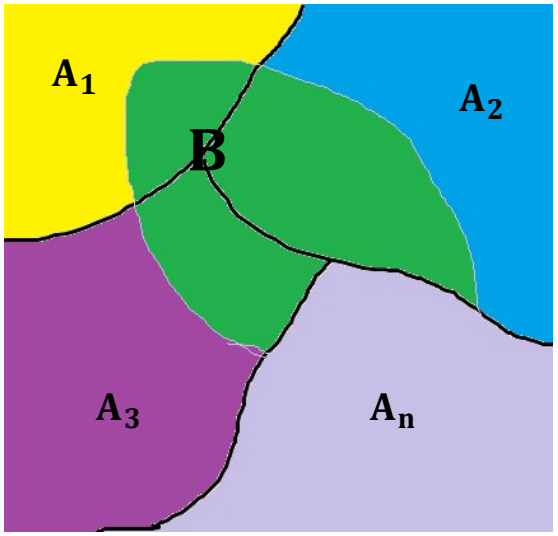
Illustrating the Rectangular Decision Zone for 2D Instances belonging to Class $\boldsymbol{\omega}$ having dimension-wise means $(\mu_1, \mu_2)$ and variances $(\sigma_1^2, \sigma_2^2)$



# Conditional Probability & Bayes Theorem

Conditional Probability: Event *A* assuming the Occurrence of *B*

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \Rightarrow P(A|B)P(B) = P(A \cap B) = P(B|A)P(A)$$




Total Probability and Bayes Theorem

$$P(B) = P(B|A_1)P(A_1) + \cdots + P(B|A_n)P(A_n)$$

Conditionally Independent Events A & B

$$P(A \cap B|\lambda) = P(A|\lambda)P(B|\lambda)$$

# Bayes Classifier

$$P(B) = P(B|A_1)P(A_1) + \cdots + P(B|A_n)P(A_n)$$

$$P(x) = P(x|\omega_1)P(\omega_1) + \cdots + P(x|\omega_n)P(\omega_n)$$

Feature Vector

Classes or Groups

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Likelihood

Prior

$$P(\omega_j|x) = \frac{P(x|\omega_j)P(\omega_j)}{P(x)}$$

Posterior

Evidence

# Bayesian Classification

Consider the classification of observed feature vector $\boldsymbol{x}$ to one of the classes $\boldsymbol{\omega_1} \dots \boldsymbol{\omega_m}$

Likelihood Prior

Posterior

$$P(\boldsymbol{\omega}_j \mid \boldsymbol{x}) = \frac{P(\boldsymbol{x} \mid \boldsymbol{\omega}_j, \boldsymbol{\theta}_j)P(\boldsymbol{\omega}_j)}{P(\boldsymbol{x})}$$

Evidence

$$P(\boldsymbol{\omega}_j \mid \boldsymbol{x}) \propto P(\boldsymbol{x} \mid \boldsymbol{\omega}_j, \boldsymbol{\theta}_j)P(\boldsymbol{\omega}_j)$$

$$\boldsymbol{x} \in \boldsymbol{\omega}_r \Rightarrow r = argmax_{j=1\dots m}\, P(\boldsymbol{\omega}_j | \boldsymbol{x})$$

# Bayes Classifier: Numerical Example

| | Class 1 $(\omega_1)$ | | | Class 2 $(\omega_2)$ | | | Class 3 $(\omega_3)$ | | |
|---|---|---|---|---|---|---|---|---|---|
| $X$ | -10.5 | -8.5 | -7.7 | 15.5 | 20.7 | 17.3 | 6.5 | 11.3 | 9.1 |

Classify: $u = 2$

Instances in Each Class are Normally Distributed

# Bayes Classifier: Numerical Example

| | Class 1 $(\omega_1)$ | | | Class 2 $(\omega_2)$ | | | Class 3 $(\omega_3)$ | | |
|---|---|---|---|---|---|---|---|---|---|
| $X$ | -10.5 | -8.5 | -7.7 | 15.5 | 20.7 | 17.3 | 6.5 | 11.3 | 9.1 |

$\mu_1 = -8.9$ $\mu_2 = 17.83$ $\mu_3 = 8.97$

$v_1 = 1.387$ $v_2 = 4.649$ $v_3 = 3.849$

$P(\omega_1) = \frac{1}{3}$ $P(\omega_2) = \frac{1}{3}$ $P(\omega_3) = \frac{1}{3}$

# Bayes Classifier: Numerical Example

| | | |
|---|---|---|
| $\mu_1 = -8.9$ | $\mu_2 = 17.83$ | $\mu_3 = 8.97$ |
| $v_1 = 1.387$ | $v_2 = 4.649$ | $v_3 = 3.849$ |
| $P(\omega_1) = \frac{1}{3}$ | $P(\omega_2) = \frac{1}{3}$ | $P(\omega_3) = \frac{1}{3}$ |

Classify: $u = 2$

$$P(\omega_j|x) = \frac{P(x|\omega_j)P(\omega_j)}{P(x)}$$

$$P(\omega_j|x) \propto P(x|\omega_j)P(\omega_j)$$

# Bayes Classifier: Numerical Example

| | | |
|---|---|---|
| $\mu_1 = -8.9$ | $\mu_2 = 17.83$ | $\mu_3 = 8.97$ |
| $v_1 = 1.387$ | $v_2 = 4.649$ | $v_3 = 3.849$ |
| $P(\omega_1) = \frac{1}{3}$ | $P(\omega_2) = \frac{1}{3}$ | $P(\omega_3) = \frac{1}{3}$ |

Classify: $u = 2$

$$P(u|\omega_1)P(\omega_1) = \left(\frac{1}{\sqrt{2\pi v_1}} e^{-\frac{1}{2v_1}(u-\mu_1)^2}\right)\left(\frac{1}{3}\right) = 2.802 \times 10^{-20}$$

# Bayes Classifier: Numerical Example

| | | |
|---|---|---|
| $\mu_1 = -8.9$ | $\mu_2 = 17.83$ | $\mu_3 = 8.97$ |
| $v_1 = 1.387$ | $v_2 = 4.649$ | $v_3 = 3.849$ |
| $P(\omega_1) = \frac{1}{3}$ | $P(\omega_2) = \frac{1}{3}$ | $P(\omega_3) = \frac{1}{3}$ |

Classify: $u = 2$

$$P(u|\omega_2)P(\omega_2) = \left(\frac{1}{\sqrt{2\pi v_2}} e^{-\frac{1}{2v_2}(u-\mu_2)^2}\right)\left(\frac{1}{3}\right) = 1.203 \times 10^{-1}$$

# Bayes Classifier: Numerical Example

| | | |
|---|---|---|
| $\mu_1 = -8.9$ | $\mu_2 = 17.83$ | $\mu_3 = 8.97$ |
| $v_1 = 1.387$ | $v_2 = 4.649$ | $v_3 = 3.849$ |
| $P(\omega_1) = \frac{1}{3}$ | $P(\omega_2) = \frac{1}{3}$ | $P(\omega_3) = \frac{1}{3}$ |

Classify: $u = 2$

$$P(u|\omega_3)P(\omega_3) = \left(\frac{1}{\sqrt{2\pi v_3}} e^{-\frac{1}{2v_3}(u-\mu_3)^2}\right)\left(\frac{1}{3}\right) = 1.239 \times 10^{-4}$$

# Bayes Classifier: Numerical Example

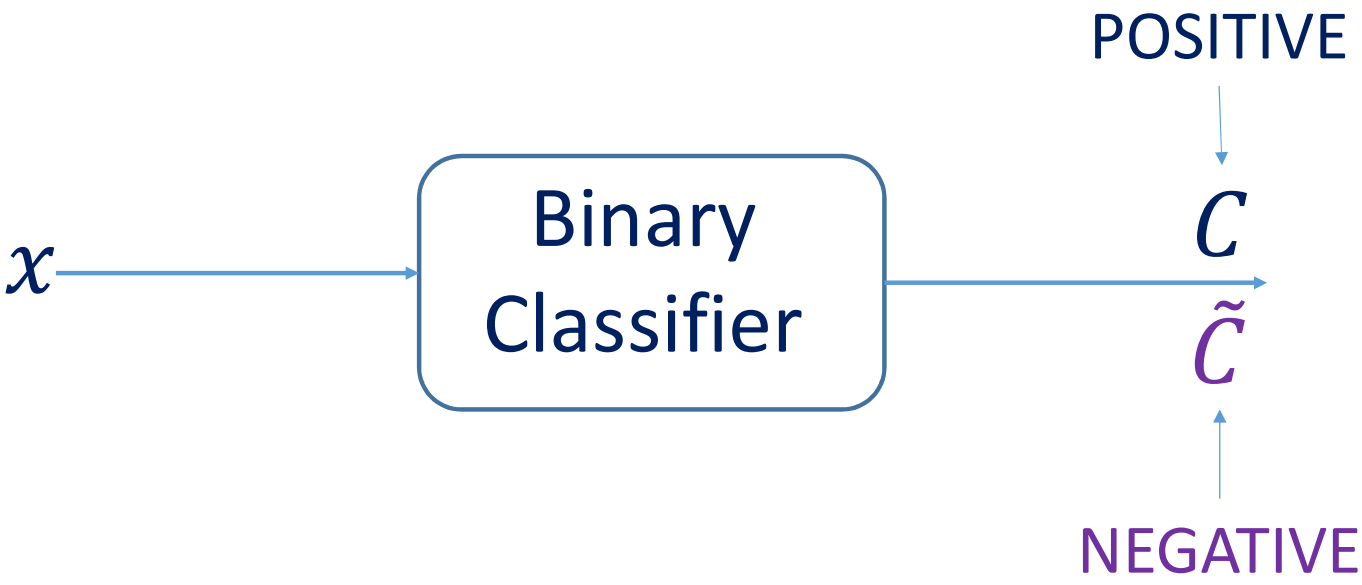
$$P(u) = P(u|\omega_1)P(\omega_1) + P(u|\omega_2)P(\omega_2) + P(u|\omega_3)P(\omega_3)$$

$$P(\omega_1|u) = \frac{P(u|\omega_1)P(\omega_1)}{P(u)} = 2.263 \times 10^{-16}$$

$$P(\omega_2|u) = \frac{P(u|\omega_2)P(\omega_2)}{P(u)} = 9.714 \times 10^{-10}$$

$$P(\omega_3|u) = \frac{P(u|\omega_3)P(\omega_3)}{P(u)} = 0.9999$$

# Binary Classification Problem



# Binary Confusion Matrix

Classification Result

Ground Truth

| | $C$ | $\tilde{C}$ |
|---|---|---|
| $C$ | TP | FN |
| $\tilde{C}$ | FP | TN |

True Positives (TP): Number of Instances from Ground Truth Positive Category Detected as Positives

True Negatives (TN): Number of Instances from Ground Truth Negative Category Detected as Negatives

False Positives (FP): Number of Instances from Ground Truth Negative Category Detected as Positives

False Negatives (FN): Number of Instances from Ground Truth Positive Category Detected as Negatives

# Binary Confusion Matrix

Classification Result

Ground Truth

| | $C$ | $\tilde{C}$ |
|---|---|---|
| $C$ | TP | FN |
| $\tilde{C}$ | FP | TN |

Ground Truth Positives (GP = TP+FN): Total Number of Ground Truth Positives

Ground Truth Positives (GN = FP+TN): Total Number of Ground Truth Negatives

Detected Positives (DP = TP+FP): Total Number of Instances Detected as Positives

Detected Negatives (DN = FN+TN): Total Number of Instances Detected as Negatives

# Binary Confusion Matrix: Example

Classification Result

Ground Truth

| | $C$ | $\tilde{C}$ | G |
|---|---|---|---|
| $C$ | 80 | 20 | 100 |
| $\tilde{C}$ | 30 | 70 | 100 |
| D | 110 | 90 | |

# Performance Measures (TPR)

Sensitivity, Recall, Hit Rate, True Positive Rate (TPR)

$$TPR = \frac{TP}{GP} = \frac{TP}{TP + FN}$$

Example

$$TPR = ?$$

# Performance Measures (TPR)

Sensitivity, Recall, Hit Rate, True Positive Rate (TPR)

$$TPR = \frac{TP}{TP + FN}$$

Example

$$TPR = \frac{80}{80 + 20} = 0.8$$

# Performance Measures (TNR)

Specificity, Selectivity, True Negative Rate (TNR)

$$TNR = \frac{TN}{GN} = \frac{TN}{TN + FP}$$

Example

$$TNR = ?$$

# Performance Measures (TNR)

Specificity, Selectivity, True Negative Rate (TNR)

$$TNR = \frac{TN}{TN + FP}$$

Example

$$TNR = \frac{70}{70 + 30} = 0.7$$

# Performance Measures (PPV)

Precision, Positive Predictive Value (PPV)

$$PPV = \frac{TP}{DP} = \frac{TP}{TP + FP}$$

Example

$$PPV = ?$$

# Performance Measures (PPV)

Precision, Positive Predictive Value (PPV)

$$PPV = \frac{TP}{TP + FP}$$

Example

$$PPV = \frac{80}{80 + 30} = 0.727$$

# Performance Measures (NPV)

Negative Predictive Value (NPV)

$$NPV = \frac{TN}{DN} = \frac{TN}{TN + FN}$$

Example

$$NPV = ?$$

# Performance Measures (NPV)

Negative Predictive Value (NPV)

$$NPV = \frac{TN}{TN + FN}$$

Example

$$NPV = \frac{70}{70 + 20} = 0.778$$

# Performance Measures (F1)

F1 Score $F1^{(p)}$ as Harmonic mean of PPV and TPR

$$F1^{(p)} = \frac{2 \times PPV \times TPR}{PPV + TPR}$$

F1 Score $F1^{(n)}$ as Harmonic mean of NPV and TNR

$$F1^{(n)} = \frac{2 \times NPV \times TNR}{NPV + TNR}$$

Example

$F1^{(p)} = ?$

$F1^{(n)} = ?$

# Performance Measures (F1)

## F1 Score (F1)

$$F1^{(p)} = \frac{2 \times PPV \times TPR}{PPV + TPR}$$

$$F1^{(n)} = \frac{2 \times NPV \times TNR}{NPV + TNR}$$

Example

$$F1^{(p)} = \frac{2 \times 0.727 \times 0.8}{0.727 + 0.8} = 0.762$$

$$F1^{(n)} = \frac{2 \times 0.778 \times 0.7}{0.778 + 0.7} = 0.737$$

# Performance Measures (FPR)

Fall Out, False Alarm Rate (FAR), False Positive Rate (FPR)

$$FPR = FAR = 1 - TNR = \frac{FP}{GN} = \frac{FP}{TN + FP}$$

Example

$$FPR = ?$$

# Performance Measures (FPR)

Fall Out, False Alarm Rate, False Positive Rate (FPR)

$$FPR = \frac{FP}{TN + FP}$$

Example

$$FPR = \frac{30}{70 + 30} = 0.3$$

# Performance Measures (ACC)

Accuracy (ACC)

$$ACC = \frac{TP + TN}{TP + FN + TN + FP}$$

Example

$$ACC = ?$$

# Performance Measures (ACC)

Accuracy (ACC)

$$ACC = \frac{TP + TN}{TP + FN + TN + FP}$$

Example

$$ACC = \frac{80 + 70}{100 + 100} = 0.75$$

# Performance Measures (BA)

Balanced Accuracy (BA)

$$BA = \frac{TPR + TNR}{2}$$

Example

$$BA = ?$$

# Performance Measures (BA)

Balanced Accuracy (BA)

$$BA = \frac{TPR + TNR}{2}$$

Example

$$BA = \frac{0.8 + 0.7}{2} = 0.75$$

# ROC Curves

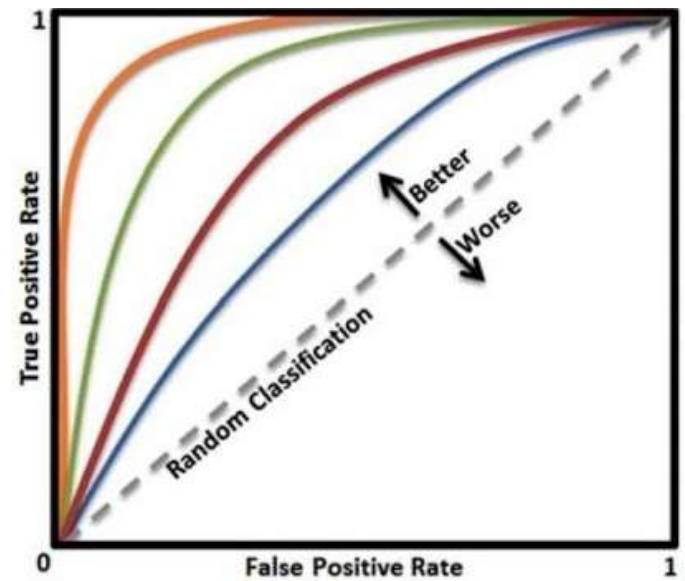
## Receiver Operating Characteristics Curves

Example $S = \{(\boldsymbol{x}_i, y_i); i = 1, \dots n\}$

Vary $\lambda$

Chebyshev Inequality based Classification

Varying TPR & FPR



# Anomaly Detection using Chebyshev Inequality

Training Data of Normal Class $\boldsymbol{\omega}$: $\boldsymbol{S} = \{\boldsymbol{x}_i; i = 1, \dots n; \boldsymbol{x} \in \mathbb{R}^{\boldsymbol{d}}\}$

$$\mu_j = \sum_{i=1}^{n} \boldsymbol{x}_i[j] \qquad j = 1, \dots d \qquad v_j = \frac{1}{n}\sum_{i=1}^{n}\left(\boldsymbol{x}_i[j] - \mu_j\right)^2$$

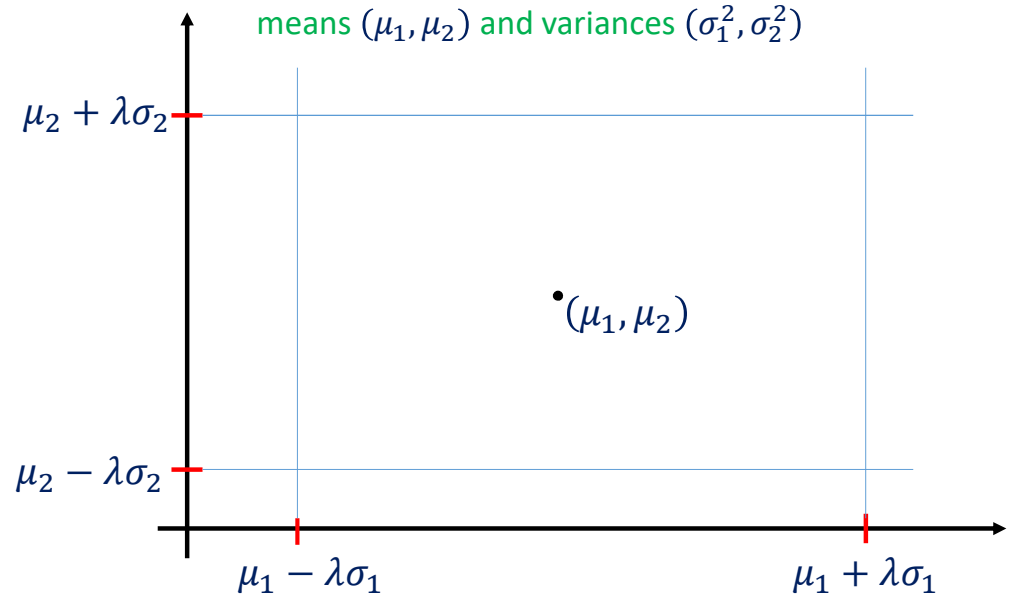
Classification Rule $\boldsymbol{R}_k$ along Dimension $k$ have Boolean Values

$$\boldsymbol{R}_k = \begin{cases} TRUE, (\boldsymbol{x}[k] - \mu_k)^2 \leq \lambda^2 v_k \\ FALSE, (\boldsymbol{x}[k] - \mu_k)^2 > \lambda^2 v_k \end{cases}$$

$$\boldsymbol{x} \in \boldsymbol{\omega} \Rightarrow \left[\bigwedge_{k=1}^{d} \boldsymbol{R}_k = TRUE\right]$$

Anomaly: Positive  Normal: Negative

Illustrating the Rectangular Decision Zone for 2D Instances belonging to Class $\boldsymbol{\omega}$ having dimension-wise means $(\mu_1, \mu_2)$ and variances $(\sigma_1^2, \sigma_2^2)$

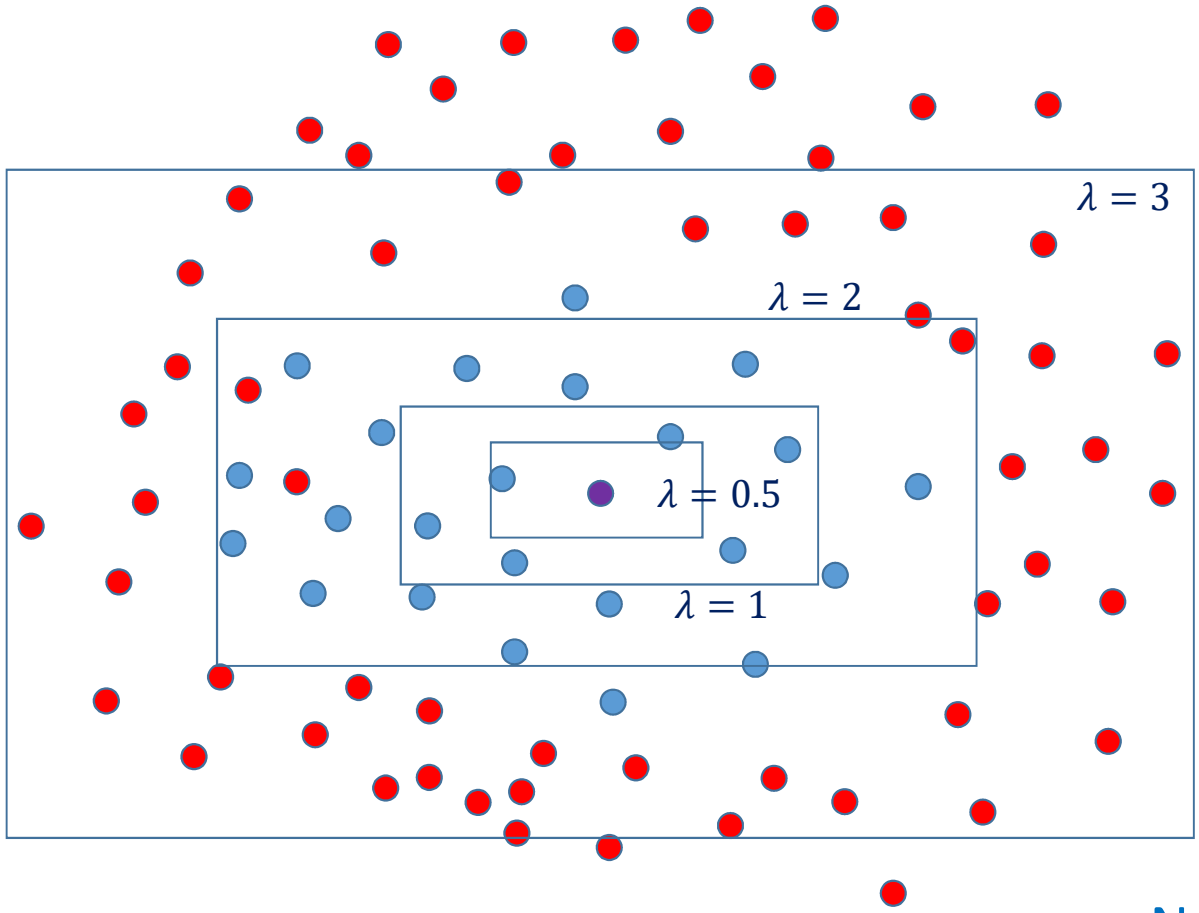


# (TPR,FPR) vs. Chebyshev Inequality Threshold

$\lambda \downarrow$

$TPR$

$FAR$




$\lambda \uparrow$

$TPR$

$FAR$

Anomaly: Positive

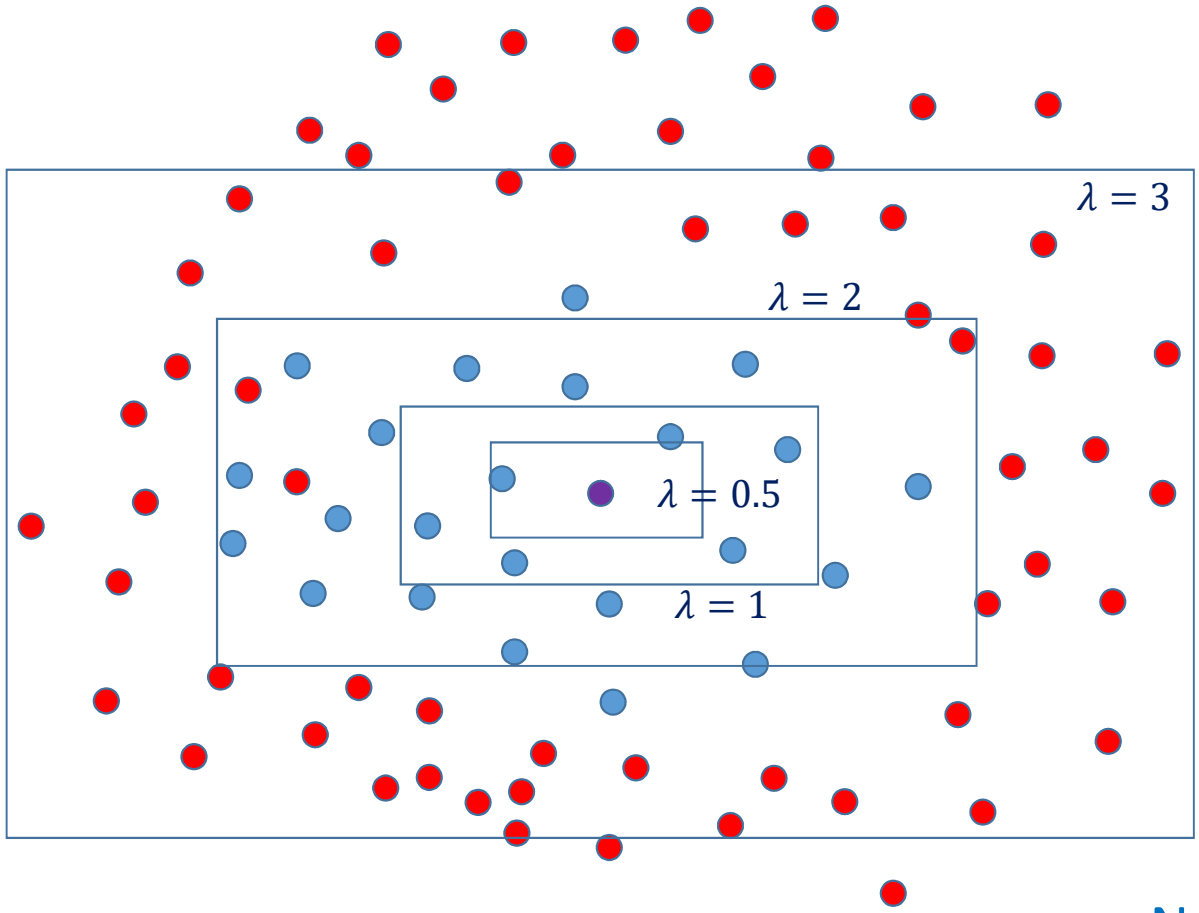
Normal: Negative

# (TPR,FPR) vs. Chebyshev Inequality Threshold

$\lambda \downarrow$

$TPR \uparrow$

$FAR \uparrow$




$\lambda \uparrow$

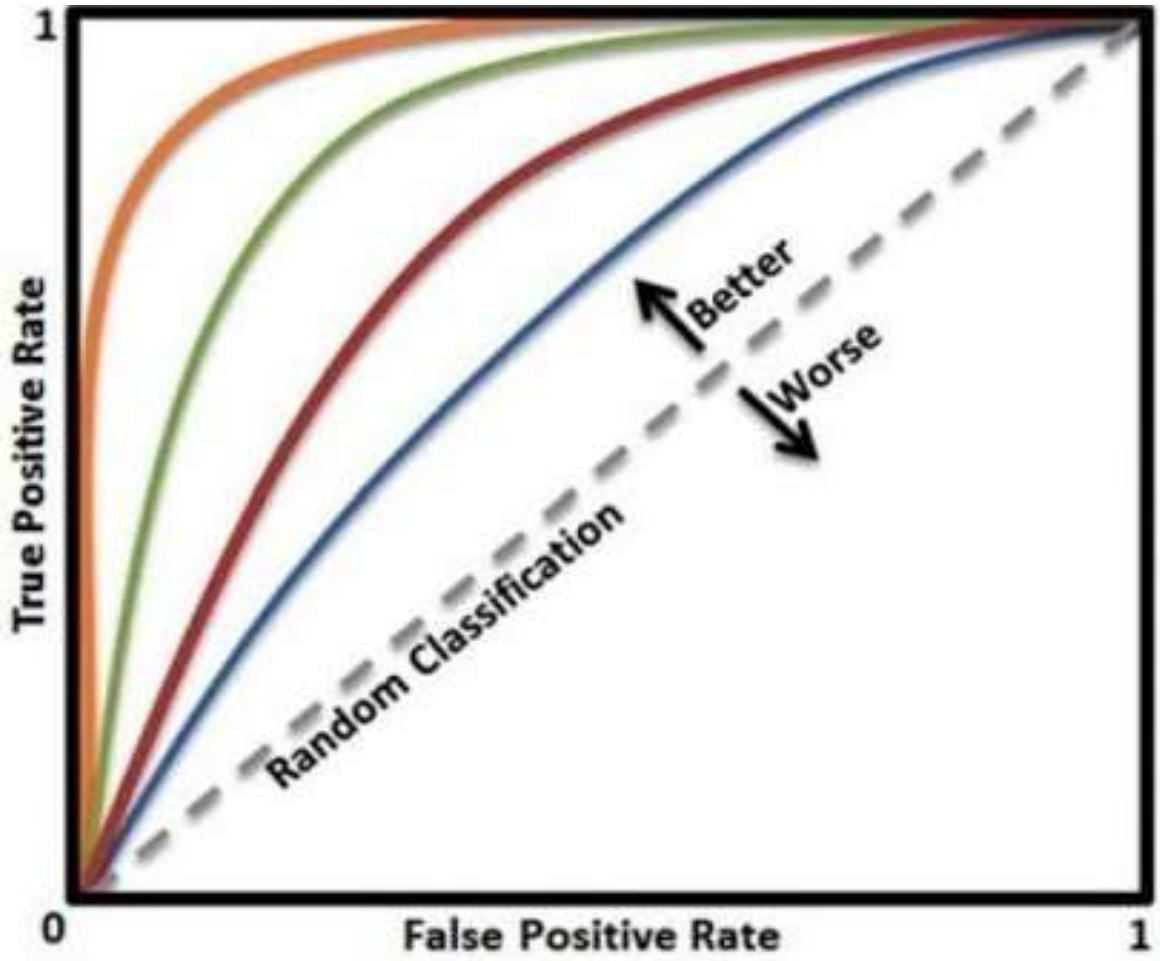
$TPR \downarrow$

$FAR \downarrow$

Anomaly: Positive

Normal: Negative

# The ROC Curve



# Multi-Category Classification Problems

$M$ Category Classification Problem $C_1, C_2, \ldots C_j, \ldots C_M$

$CM$ : Confusion Matrix

$CM[i][j]$ : Entity of $C_i$ Detected as that of $C_j$

Example

| $C_1$ | $C_2$ | $C_3$ | $C_4$ | $C_5$ |
|---|---|---|---|---|
| 100 | 100 | 100 | 100 | 100 |

# Confusion Matrix

Classification Result

Ground Truth

| | $C_1$ | $C_2$ | $C_3$ | $C_4$ | $C_5$ | G |
|---|---|---|---|---|---|---|
| $C_1$ | 60 | 25 | 0 | 5 | 10 | 100 |
| $C_2$ | 10 | 50 | 10 | 20 | 10 | 100 |
| $C_3$ | 5 | 5 | 80 | 0 | 10 | 100 |
| $C_4$ | 40 | 20 | 10 | 30 | 0 | 100 |
| $C_5$ | 20 | 50 | 10 | 0 | 20 | 100 |
| D | 135 | 150 | 110 | 55 | 50 | |

# Performance Measures (ACC)

Overall Accuracy (ACC)

$$ACC = \frac{\sum_{j=1}^{M} CM[j][j]}{\sum_{i=1}^{M} \sum_{j=1}^{M} CM[i][j]}$$

Example

$$ACC = ?$$

# Performance Measures (ACC)

Overall Accuracy (ACC)

$$ACC = \frac{\sum_{j=1}^{M} CM[j][j]}{\sum_{i=1}^{M} \sum_{j=1}^{M} CM[i][j]}$$

Example

$$ACC = \frac{60 + 50 + 80 + 30 + 20}{100 + 100 + 100 + 100 + 100} = 0.48$$

# Class-wise F1 Score

| | $C_1$ | $\cdots$ | $\cdots$ | $C_j$ | $\cdots$ | $C_m$ |
|---|---|---|---|---|---|---|
| $C_1$ | | | | | | |
| $\vdots$ | | | | | | |
| $C_i$ | | | | | | |
| $\vdots$ | | | | | | |
| $C_m$ | | | | | | |

$CM[i][j]$ : Entity of $C_i$
Detected as that of $C_j$

$$PPV[i] = ? \qquad TPR[i] = ?$$

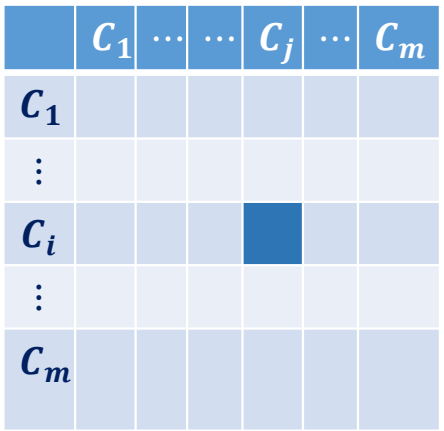
$$F1[i] = ?$$

Example

$$PPV[4] = ? \qquad TPR[4] = ?$$

$$F1[4] = ?$$

# Class-wise F1 Score



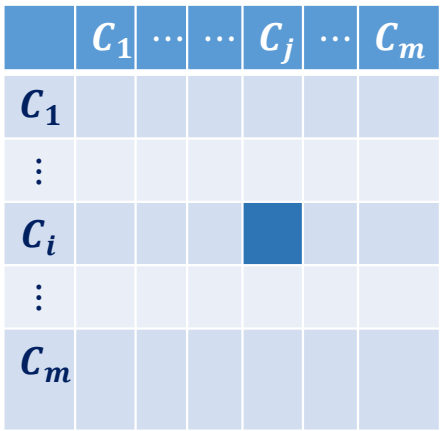

$CM[i][j]$ : Entity of $C_i$
Detected as that of $C_j$

$DP$ : Detected Positives

$GP$ : Ground-Truth Positives

$$PPV[i] = \frac{TP[i]}{DP[i]} = \frac{CM[i][i]}{\sum_{k=1}^{m} C[k][i]}$$

# Class-wise F1 Score



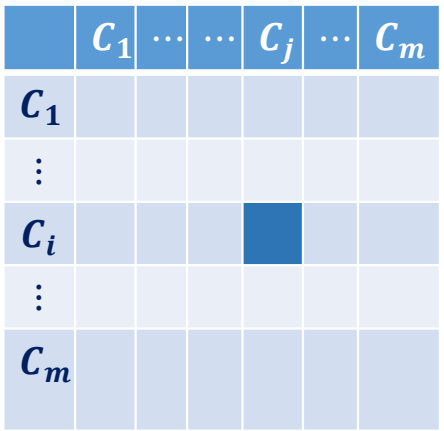

$CM[i][j]$ : Entity of $C_i$
Detected as that of $C_j$

$DP$ : Detected Positives

$GP$ : Ground-Truth Positives

$$TPR[i] = \frac{TP[i]}{GP[i]} = \frac{CM[i][i]}{\sum_{k=1}^{m} C[i][k]}$$

# Class-wise F1 Score




$CM[i][j]$ : Entity of $C_i$ Detected as that of $C_j$

$DP$ : Detected Positives

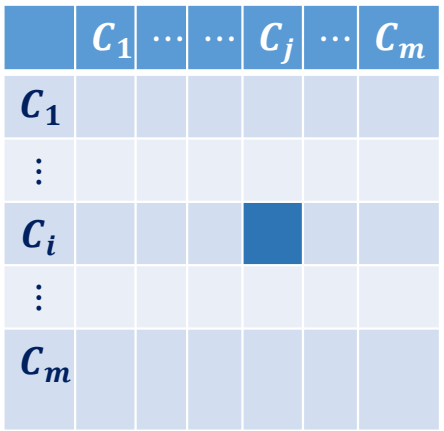
$GP$ : Ground-Truth Positives

$$PPV[i] = \frac{TP[i]}{DP[i]} = \frac{CM[i][i]}{\sum_{k=1}^{m} C[k][i]}$$

$$TPR[i] = \frac{TP[i]}{GP[i]} = \frac{CM[i][i]}{\sum_{k=1}^{m} C[i][k]}$$

$$F1[i] = \frac{2 \times PPV[i] \times TPR[i]}{PPV[i] + TPR[i]}$$

# Class-wise F1 Score




$CM[i][j]$ : Entity of $C_i$
Detected as that of $C_j$

$DP$ : Detected Positives

$GP$ : Ground-Truth Positives

$$PPV[4] = \frac{CM[4][4]}{\sum_{k=1}^{5} C[k][4]} = \frac{30}{55} = 0.5455$$

$$TPR[4] = \frac{CM[4][4]}{\sum_{k=1}^{5} C[4][k]} = \frac{30}{100} = 0.3$$

$$F1[4] = \frac{2 \times 0.5455 \times 0.3}{0.5455 + 0.3} = 0.3871$$

# Reporting Performances

Total Training Data Set

| | | | | |
|---|---|---|---|---|
| $D_1$ | $D_2$ | $D_3$ | $D_4$ | $D_5$ |

Report: $\mu(A), \sigma(A)$

| Train | Validate | Performance |
|---|---|---|
| $D_1, D_2, D_3, D_4,$ | $D_5$ | $A_1$ |
| $D_2, D_3, D_4, D_5,$ | $D_1$ | $A_2$ |
| $D_3, D_4, D_5, D_1,$ | $D_2$ | $A_3$ |
| $D_4, D_5, D_1, D_2,$ | $D_3$ | $A_4$ |
| $D_5, D_1, D_2, D_3,$ | $D_4$ | $A_5$ |

# Model Selection

Four machine learning models are trained on the same dataset and are tested across five different datasets (representing different scenarios). The accuracies (in percentage) of each model are tabulated for these datasets.

| | Dataset #1 | Dataset #2 | Dataset #3 | Dataset #4 | Dataset #5 |
|---|---|---|---|---|---|
| Model A | 50 | 40 | 70 | 30 | 80 |
| Model B | 90 | 92 | 89 | 91 | 90 |
| Model C | 60 | 61 | 59 | 58 | 62 |
| Model D | 95 | 70 | 85 | 99 | 60 |

Rank the Models based on *Performance* and *Consistency* by using their *Accuracy Statistics*.

# Model Selection

| | D1 | D2 | D3 | D4 | D5 | M | SD |
|---|---|---|---|---|---|---|---|
| Model A | 50 | 40 | 70 | 30 | 80 | 54 | 18.5472 |
| Model B | 90 | 92 | 89 | 91 | 90 | 90.4 | 1.0198 |
| Model C | 60 | 61 | 59 | 58 | 62 | 60 | 1.4142 |
| Model D | 95 | 70 | 85 | 99 | 60 | 81.8 | 14.7973 |

Ranked by Performance: B > D > C > A

Ranked by Consistency: B > C > D > A

Best Model: B

Worst Model: A

# Summary

- Chebyshev Inequality as Classifier
- Bayesian Classification and the MAP Rule
- Binary Confusion Matrix
- Performance Measures
- The ROC Curve
- Confusion Matrix for Multi-Category Classification
- Reporting Performance Measures

# Thank You